RaterBench: How Reliable Are Small Open LLMs as Expert Text Annotators?

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Abstract

More and more researchers ask large language models (LLMs) to label text that trained annotators used to label. I test how far that can be trusted on two public social-media tasks: stress in Reddit posts (Dreaddit) and stance in tweets (TweetEval). Two open 7–8B models, Llama 3.1 and Qwen 2.5, labeled every item under four prompting strategies, and I compared them with a fine-tuned RoBERTa model using the statistics normally used for inter-rater reliability, with bootstrap confidence intervals. With its best prompt, each LLM agrees with the gold labels about as well as the fine-tuned model does (Cohen's κ 0.64 vs. 0.64 on Dreaddit, 0.56 vs. 0.49 on TweetEval). Zero-shot prompting falls well short, and the better model depends on the task. Of three confidence signals, only token probabilities separate right from wrong answers usefully. Split conformal triage built on them sends 15% of Dreaddit items to a human at a 90% coverage target, but 56% of stance items. Labels also change under 6.9–9.0% of meaning-preserving paraphrases. The code, prompts and fixed sample IDs are public.

1 Introduction

Labeling text by hand is slow and expensive, so it is no surprise that researchers in psychiatry, public health and computational social science have started handing the job to LLMs. The worry is that the models make mistakes quietly and systematically. Before relying on LLM labels, a researcher would want to know four things. How closely do they agree with expert labels (RQ1)? Does the model's confidence say which labels to trust (RQ2)? Can that confidence decide which items still need a human, with a guaranteed coverage level (RQ3)? And do the errors cluster in particular subgroups, or change when the wording changes but the meaning does not (RQ4)?

RaterBench answers these with the tools used to judge human annotators: Cohen's κ [3] and Krippendorff's α [6] with 95% bootstrap intervals, paired McNemar tests [8] with Holm correction [5], standard calibration metrics [4], and split conformal prediction [1]. Every number in this report is produced by scripts in the repository from cached model outputs.

2 Related work

[To be written: LLMs as annotators in the social sciences; uncertainty and calibration of LLMs; conformal prediction for classification; robustness and bias audits of text classifiers.]

3 Data and setup

Datasets. *Dreaddit* [12] contains excerpts of Reddit posts from ten subreddits (grouped into five domains), labeled stress or not stress by crowd workers, whose own agreement was moderate (Fleiss' $\kappa = 0.47$ in the original paper). *TweetEval stance* [2], which comes from SemEval-2016 Task 6 [9], labels tweets as favoring, opposing, or taking no stance on one of five targets. I evaluate on fixed stratified samples of the test splits (715 Dreaddit and 1,000 TweetEval items). Conformal prediction gets its own calibration set that never overlaps the evaluation set: the remaining 249 TweetEval test items, and 305 items held out from Dreaddit's training split, because its test split was too small to divide. Few-shot examples are drawn only from training data.

Models and prompts. Llama 3.1 8B Instruct and Qwen 2.5 7B Instruct, both 4-bit and served by Ollama, labeled every item under four prompts. *Zero-shot* gives the task and the label definitions. *Few-shot* adds five label-balanced training examples. *Codebook* replaces them with a detailed annotation guideline covering inclusion rules and edge cases, and *codebook+reason* also asks for a short rationale before the label. The models answer in JSON with a label and a 0–100 confidence, and a malformed answer gets one corrective retry. Main runs use temperature 0. Producing all 40,842 labels took 3.6 hours on one laptop GPU, and only 1 answer could not be parsed (it counts as an error). The baselines are the majority class and RoBERTa-base [7] fine-tuned on the training split. I trained it with three seeds and report seed 13 with bootstrap intervals; across seeds its κ is 0.62 $\pm$ 0.02 on Dreaddit and 0.48 $\pm$ 0.01 on TweetEval.

Confidence signals. With the codebook prompt I compare three signals: *verbalized* confidence, the number the model reports; *token probability*, the product of the probabilities of the tokens that spell the label; and *self-*

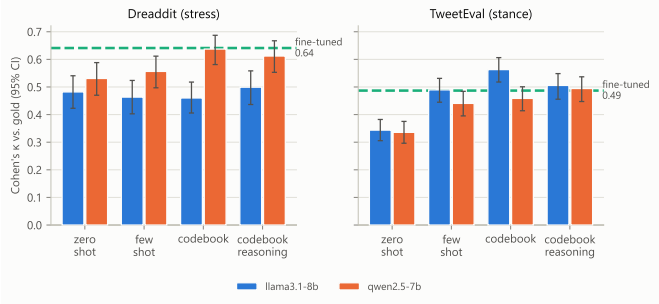


Figure 1: Cohen's κ by prompting strategy (95% CIs). Dashed line: fine-tuned RoBERTa.

consistency [14], the share of five extra answers sampled at temperature 0.7 that agree with the temperature-0 label.

Conformal triage. On the calibration set I fit split-conformal prediction sets with the LAC score [11]; adaptive prediction sets [10] are in the repository too. An item whose set contains exactly one label is labeled automatically, and everything else goes to a human. The LLMs only give a probability for the label they chose, so the rest of the probability is spread evenly over the other labels.

4 Results

RQ1: Agreement. The prompt turns out to matter about as much as the model (Table 1, Figure 1). With the right prompt, each LLM catches up with the fine-tuned baseline on one task: Qwen 2.5 7B with the codebook on Dreaddit (accuracy 81.8% against 82.1%, Holm-adjusted McNemar $p = 1.00$), and Llama 3.1 8B with the codebook on TweetEval (73.9% against 69.4%, $p = 0.14$). Zero-shot prompting is significantly worse than the fine-tuned model for both models on both tasks (largest adjusted $p = 0.046$). Stance shows the gap most clearly: zero-shot κ is only 0.34, and the codebook improves both models ($p < 0.001$). The two models are good at different things. With the codebook, Qwen is better on stress ($p < 0.001$) and Llama on stance ($p < 0.001$). Asking for reasoning first helped in two of the four model–task pairs and hurt in the other two.

RQ2: Calibration. Table 2 compares the three signals. Token probability ranks right and wrong answers most consistently (AUROC 0.69–0.80), even though its values are poorly calibrated in absolute terms. Verbalized confidence is coarse: no model used more than 6 distinct values on a task, and its AUROC ranges from 0.52, barely above chance, to 0.77. A low ECE can hide this. Qwen's verbalized confidence on Dreaddit is well calibrated on average but hardly distinguishes its errors. Self-consistency helps even less (AUROC 0.54–0.58): at temperature 0.7 the extra samples almost always repeat the first answer, so the models seem to make their mistakes consistently rather than at random.

RQ3: Conformal triage. At a 90% coverage target (Table 3), Qwen 2.5 7B's token probabilities send 15% of Dreaddit items to a human. The items it labels automatically have 13.0% errors, against 18.2% if it labeled everything, which is close to what the fine-tuned model achieves (16% routed, 13.5% errors). Stance is much harder: even the best LLM sends 56% of items to a human. Coarse signals make triage pointless. Self-consistency sends 100% of items to a human in every setting, and verbalized confidence at least 88% on stance. Empirical coverage stayed within the expected finite-sample range in all 56 of 56 settings [13]. It is worth being clear about what is guaranteed, though: the coverage of the *set*, not the error rate among auto-labeled items, which can be higher than the miscoverage level.

RQ4: Subgroups and robustness. On Dreaddit, both LLMs make errors at different rates across domains (χ^2 test, $p < 0.001$ for Llama and $p = 0.020$ for Qwen, with macro-F1 gaps of up to 0.18 and 0.16), while the fine-tuned model does not significantly ($p = 0.069$). TweetEval shows the opposite pattern. There the fine-tuned model's errors vary across targets ($p < 0.001$) and the LLMs' less clearly ($p = 0.11$, $p = 0.050$; Figure 2). These groups are topics and communities, not demographic groups, and they differ in size and label mix. Edits that keep the meaning still change labels (Table 4). Across both models and the zero-shot and codebook prompts, labels flip for 1.7–3.3% of gender-swapped Dreaddit posts, 2.9–8.0% of items with typos, and 6.9–9.0% of paraphrases.

5 Failure modes

To see what goes wrong, I sampled 100 disagreements between model and gold (25 per model and task, from the codebook+reason prompt so the model's rationale is visible) and gave each one a primary error type from a ten-type taxonomy. The codes were drafted with a language model and reviewed by the author; the codes, the reason for each, and the taxonomy are in the repository. Table 5 has the counts.

The most common type on Dreaddit is *label boundary* (20 of 50). The codebook asks about the writer's *current* state, but the crowd often labeled by topic: a calm account of past abuse was labeled stress, and a crisis described matter-of-factly was labeled not stress. The models followed the codebook and ended up disagreeing with the crowd. Another 13 Dreaddit errors look like mistakes in the gold labels, such as a post where the writer says they feel "absolutely awful" labeled not stress. On TweetEval the most common type is *ambiguous gold* (14 of 50); several tweets labeled *against* legal abortion are plainly pro-choice, for instance backing the repeal of Ireland's abortion ban. The genuine model errors there are reading too much into weak cues (8), missing sarcasm (7), missing implied stances (7) and confusing the target with a related entity (5). Twice, a model even inverted the codebook's

Model	Prompt	Dreaddit (stress)		TweetEval (stance)	
		κ [95% CI]	F1	κ [95% CI]	F1
Majority class	–	0.00 [0.00, 0.00]	0.34	0.00 [0.00, 0.00]	0.24
RoBERTa-base (fine-tuned)	–	0.64 [0.58, 0.69]	0.82	0.49 [0.44, 0.53]	0.66
Llama 3.1 8B	zero-shot	0.48 [0.42, 0.54]	0.73	0.34 [0.31, 0.38]	0.54
	few-shot	0.46 [0.40, 0.52]	0.72	0.49 [0.44, 0.53]	0.66
	codebook	0.46 [0.41, 0.52]	0.72	**0.56** [0.52, 0.61]	0.71
	codebook+reason	0.50 [0.44, 0.56]	0.74	0.51 [0.46, 0.55]	0.65
Qwen 2.5 7B	zero-shot	0.53 [0.47, 0.59]	0.76	0.34 [0.30, 0.38]	0.53
	few-shot	0.56 [0.50, 0.61]	0.77	0.44 [0.40, 0.48]	0.63
	codebook	**0.64** [0.58, 0.69]	0.82	0.46 [0.41, 0.50]	0.64
	codebook+reason	0.61 [0.55, 0.67]	0.81	0.49 [0.45, 0.54]	0.66

Table 1: Agreement with gold labels on the evaluation sets (Cohen's κ with 95% bootstrap CIs over items; macro-F1). Bold: best LLM configuration per dataset. Krippendorff's α tracks κ closely and is reported in the repository.

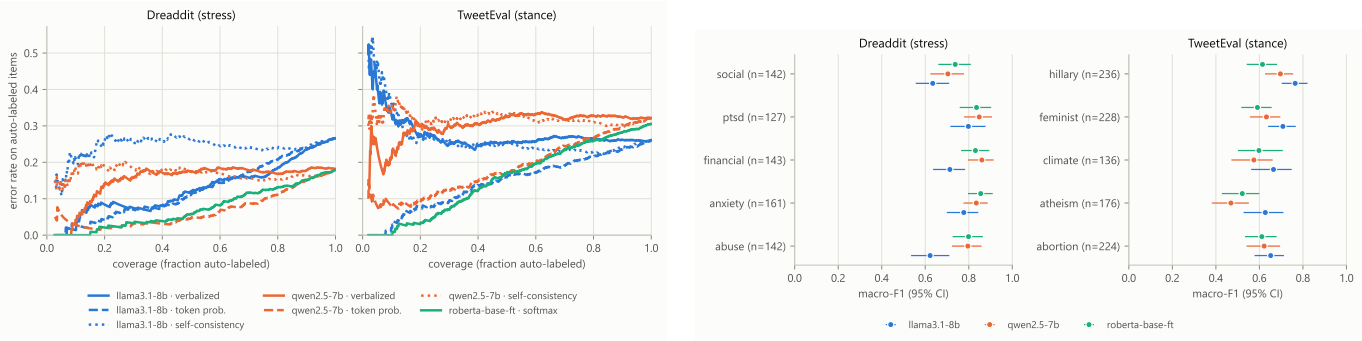


Figure 2: Left: risk–coverage curves (error among auto-labeled items as more items are auto-labeled, most confident first). Right: macro-F1 per Dreaddit domain and TweetEval target (codebook prompt, 95% CIs).

Model	Signal	Dreaddit		TweetEval	
		ECE↓	AUROC↑	ECE↓	AUROC↑
Llama 3.1 8B	verbalized	0.17	0.77	0.11	0.52
	token prob.	0.22	0.75	0.17	0.69
	self-consistency	0.24	0.56	0.22	0.58
Qwen 2.5 7B	verbalized	0.05	0.55	0.18	0.53
	token prob.	0.16	0.80	0.29	0.71
	self-consistency	0.17	0.56	0.30	0.54
RoBERTa-base	softmax	0.14	0.76	0.17	0.72

Table 2: Calibration of confidence signals (codebook prompt, eval sets).

Model	Signal	Dreaddit			TweetEval		
		Cov.	Human%	Err.%	Cov.	Human%	Err.%
Llama 3.1 8B	token prob.	0.890	33	16.4	0.939	56	14.0
	verbalized	0.968	60	8.1	0.963	88	30.6
	self-consistency	1.000	100	–	1.000	100	–
Qwen 2.5 7B	token prob.	0.890	15	13.0	0.943	61	14.7
	verbalized	1.000	92	0.0	0.988	93	16.9
	self-consistency	1.000	100	–	1.000	100	–
RoBERTa-base	softmax	0.887	16	13.5	0.927	60	12.6

Table 3: LAC conformal triage at 90% target coverage (codebook prompt). Cov.: empirical coverage; Human%: items routed to a human; Err.%: error on auto-labeled items.

rule that criticizing religion means favoring atheism. In total, 27 of the 100 errors point at the gold label rather than the model. Agreement with gold therefore somewhat understates how good the LLMs are, and the same holds for the fine-tuned baseline, which learned from those labels.

6 Limitations

I evaluated only two small open models, and larger or proprietary models may behave differently. The gold labels are noisy themselves (Dreaddit's annotators only agreed moderately), so a disagreement is not always the model's fault. Because Dreaddit's calibration items come from its training split, the conformal guarantee is only approximate there. The subgroups are topics and communities rather than demographic groups, which limits the bias audit, and I did not correct the subgroup tests for multiple comparisons. The paraphrases were written by Qwen, which is also one of the models under test, and it sometimes toned down emotional language. The LLM confidence signals cover only the chosen label, so the conformal sets spread the rest of the probability evenly. Each LLM result is a single run at temperature 0. Finally, the error taxonomy has a single coder: the codes were drafted with a language model and reviewed by the author.

7 Ethics

Both datasets are public social-media text released for research. I do not redistribute any of the text, only item IDs and hashes. The project uses no clinical data and makes

Model	Edit	Dreaddit	TweetEval
Llama 3.1 8B	gender swap	1.7 [0.0, 4.2]	n/a
	typo	5.3 [2.4, 8.3]	6.5 [3.5, 10.0]
	paraphrase	8.4 [4.9, 12.3]	8.5 [5.0, 12.5]
Qwen 2.5 7B	gender swap	2.5 [0.0, 5.8]	n/a
	typo	2.9 [1.0, 5.3]	5.5 [2.5, 9.0]
	paraphrase	6.9 [3.9, 10.3]	7.5 [4.0, 11.5]

Table 4: Label flip rate (%, 95% CI) under edits, codebook prompt, 200-item subsets. Gender swaps apply only to Dreaddit posts containing gendered words.

Error type	Dreaddit	TweetEval	Total
ambiguous gold	13	14	27
label boundary	20	4	24
implicit stance	8	7	15
over inference	5	8	13
sarcasm irony	0	7	7
target confusion	0	5	5
keyword reliance	0	3	3
negation	2	0	2
missing context	0	2	2
other	2	0	2

Table 5: Primary error type in 100 sampled LLM errors (codes drafted with a language model and reviewed by the author).

no clinical claims; an LLM labeling a post as stressed is not a mental-health assessment. Neither dataset comes with an explicit license, so I follow the original sources' terms and the terms of service of Reddit and Twitter.

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